

Personal info

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Introduction: This SQL exercise shows the extractions of dates (day, Month, Year), DATEDIFF and CASE statements queries.

Q1: The purpose of this code is to show the day name

```
SELECT
    orders AS order_id,
    _c1 AS customer_id,
    _c2 AS order_date,
    DAYNAME(TO_DATE(_c2, 'M/d/yy')) AS day_name
FROM `retail`.`default`.`orders`
WHERE orders != 'order_id';
```

Add parameter

← Results 2 of 2 →

Table ▾ +

	^B _C order_id	^B _C customer_id	^B _C order_date	^B _C day_name
1	1001	101	5/1/26	Fri
2	1002	102	5/2/26	Sat
3	1003	103	5/3/26	Sun
4	1004	104	5/4/26	Mon
5	1005	105	5/5/26	Tue

Q2: the syntax shows the signup month name

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```
select `Month name` AS customer_id,  
       _c1 AS customer_name,  
       _c2 AS signup_date,  
       MONTHNAME(TO_DATE(_c2, 'yyyy-MM-dd')) AS signup_month_name  
FROM `retail`.`default`.`month_name`  
WHERE `Month name` != 'customer_id '  
LIMIT 100;
```

Add parameter

	customer_id	customer_name	signup_date	signup_month_name
1	201	John	2026-01-15	Jan
2	202	Mary	2026-02-20	Feb
3	203	Peter	2026-03-05	Mar
4	204	Sarah	2026-04-18	Apr
5	205	Thabo	2026-05-30	May

Q3: Is about show the month number from the table.

```
SELECT  
  `Month number from sale date` AS sale_id,  
  _c1 AS product_name,  
  _c2 AS sale_date,  
  _c3 AS amount,  
  MONTH(TO_DATE(_c2, 'M/d/yy')) AS month_number  
FROM `retail`.`default`.`month_number_from_sale_date`  
WHERE `Month number from sale date` != 'sale_id';
```

Add parameter

	sale_id	product_name	sale_date	amount	month_number
1	501	Laptop	1/10/26	12000	1
2	502	Mouse	2/15/26	250	2
3	503	Keyboard	3/20/26	700	3
4	504	Monitor	4/25/26	3500	4
5	505	Desk	5/30/26	2000	5

Q4: The syntax written here is to show the transaction year.

```
SELECT
  `Transaction` AS transaction_id,
  _c1 AS customer_id,
  _c2 AS transaction_date,
  YEAR(TO_DATE(_c2, 'M/d/yy')) AS transaction_year
FROM `retail`.`default`.`transaction`
WHERE `Transaction` != 'Transaction_id'
```

Add parameter

Table ▾ + 🔍 🏠

	^A _C transaction_id	^A _C customer_id	^A _C transaction_date	¹ ₃ transaction_year
1	T001	101	12/15/24	2024
2	T002	102	1/20/25	2025
3	T003	103	6/10/25	2025
4	T004	104	2/5/26	2026
5	T005	105	5/25/26	2026

Q5: The purpose of this syntax is to extract the day of the month.

```
SELECT
  `Day of the month` AS delivery_id,
  _c1 AS customer_id,
  _c2 AS delivery_date,
  DAY(TO_DATE(_c2, 'M/d/yy')) AS day_of_month
FROM `retail`.`default`.`day_of_month`
WHERE `Day of the month` != 'delivery_id'
```

Add parameter

Table ▾ + 🔍

	^A _C delivery_id	^A _C customer_id	^A _C delivery_date	¹ ₃ day_of_month
1	D001	101	5/1/26	1
2	D002	102	5/8/26	8
3	D003	103	5/15/26	15
4	D004	104	5/22/26	22
5	D005	105	5/29/26	29

Q6: The purpose of this syntax is to show today's date.

```
SELECT
  `Employees` AS employee_id,
  _c1 AS employee_name,
  _c2 AS department,
  CURRENT_DATE() AS today_date
FROM `retail`.`default`.`employees`
WHERE `Employees` != 'employee_id'
```

Add parameter

	employee_id	employee_name	department	today_date
1	301	Nandi	Sales	2026-06-07
2	302	Brian	IT	2026-06-07
3	303	Lerato	Finance	2026-06-07
4	304	Sipho	HR	2026-06-07
5	305	Aisha	Marketing	2026-06-07

Q7: The purpose of this syntax is to show order date.

```
SELECT
  `Online orders` AS order_id,
  _c1 AS customer_id,
  _c2 AS order_date_text,
  TO_DATE(_c2, 'M/d/yy') AS order_date
FROM `retail`.`default`.`online_orders`
WHERE `Online orders` != 'order_id'
```

Add parameter

	order_id	customer_id	order_date_text	order_date
1	4001	101	1/15/26	2026-01-15
2	4002	102	2/20/26	2026-02-20
3	400	103	3/25/26	2026-03-25
4	4004	104	4/10/26	2026-04-10
5	4005	105	5/5/26	2026-05-05

Q8 The purpose of this syntax is to show formatted payment date.

```
SELECT
  `format date as text` AS payment_id,
  _c1 AS customer_id,
  _c2 AS payment_date,
  DATE_FORMAT(TO_DATE(_c2, 'M/d/yy'), 'yyyy-MM-dd') AS formatted_payment_date
FROM `retail`.`default`.`format_date_as_text`
WHERE `format date as text` != 'payment_id'
```

Add parameter

	^A _C payment_id	^A _C customer_id	^A _C payment_date	^A _C formatted_payment_date
1	P001	101	1/5/26	2026-01-05
2	P002	102	2/10/26	2026-02-10
3	P003	103	3/15/26	2026-03-15
4	P004	104	4/20/26	2026-04-20
5	P005	105	5/25/26	2026-05-25

Q9: This syntax is to extract days since last purchase.

```
SELECT
  `Days since last purchase` AS customer_id,
  _c1 AS customer_name,
  _c2 AS last_purchase_date,
  DATEDIFF(CURRENT_DATE(), TO_DATE(_c2, 'M/d/yy')) AS days_since_last_purchase
FROM `retail`.`default`.`days_since_last_purchase`
WHERE `Days since last purchase` != 'customer_id'
```

Add parameter

	^A _C customer_id	^A _C customer_name	^A _C last_purchase_date	¹ ₂ ³ days_since_last_purchase
1	501	John	5/1/26	37
2	502	Mary	5/10/26	28
3	503	Peter	5/15/26	23
4	504	Sarah	5/20/26	18
5	505	Thabo	5/25/26	13

Q10 This syntax is to extract the expected delivery date.

```
SELECT
  `shipping orders` AS order_id,
  _c1 AS customer_id,
  _c2 AS order_date,
  DATE_ADD(TO_DATE(_c2, 'M/d/yy'), 7) AS expected_delivery_date
FROM `retail`.`default`.`shipping_orders`
WHERE `shipping orders` != 'order_id'
```

Add parameter

	^A _C order_id	^A _C customer_id	^A _C order_date	expected_delivery_date
1	6001	101	5/1/26	2026-05-08
2	6002	102	5/3/26	2026-05-10
3	6003	103	5/5/26	2026-05-12
4	6004	104	5/7/26	2026-05-14
5	6005	105	5/9/26	2026-05-16

Q11: This syntax is for extracting booking dates from a day, Month and year.

```
SELECT
  `Bookings` AS booking_id,
  _c1 AS customer_id,
  TO_DATE(_c2, 'M/d/yy') AS booking_date,
  YEAR(TO_DATE(_c2, 'M/d/yy')) AS booking_year,
  MONTH(TO_DATE(_c2, 'M/d/yy')) AS booking_month,
  DAY(TO_DATE(_c2, 'M/d/yy')) AS booking_day
FROM `retail`.`default`.`bookings`
WHERE `Bookings` != 'Booking_id'
```

Add parameter

	ing_id	^A _C customer_id	booking_date	¹ ₃ booking_year	¹ ₃ booking_month	¹ ₃ booking_day
1		101	2026-01-12	2026	1	12
2		102	2026-02-18	2026	2	18
3		103	2026-03-22	2026	3	22
4		104	2026-04-09	2026	4	9
5		105	2026-05-27	2026	5	27

Q12 This syntax is for extracting year orders that only shows the year 2026.

```
SELECT
  `yearly_orders` AS order_id,
  _c1 AS customer_id,
  TO_DATE(_c2, 'M/d/yy') AS order_date,
  YEAR(TO_DATE(_c2, 'M/d/yy')) AS order_year,
  _c3 AS amount
FROM `retail`.`default`.`yearly_orders`
WHERE `yearly_orders` != 'Order_id'
AND YEAR(TO_DATE(_c2, 'M/d/yy')) = 2026;
```

Add parameter

	order_id	customer_id	order_date	order_year	amount
1	7004	104	2026-02-05	2026	1500
2	7005	105	2026-05-25	2026	2000

Q13 This syntax extracts the orders made on the month of month

```
SELECT
  _c0 AS order_id,
  _c1 AS customer_id,
  TO_DATE(_c2, 'M/d/yy') AS order_date,
  MONTH(TO_DATE(_c2, 'M/d/yy')) AS order_month,
  _c3 AS amount
FROM `retail`.`default`.`march_orders`
WHERE _c0 != 'Order_id'
AND MONTH(TO_DATE(_c2, 'M/d/yy')) = 3;
```

Add parameter

	order_id	customer_id	order_date	order_month	amount
1	3	103	2026-03-10	3	800

Q14 This syntax extracts the last day of the month

```
SELECT
  subscription_id AS subscription_id,
  customer_id,
  `update` AS start_date,
  LAST_DAY(`update`) AS month_end_date
FROM `retail`.`default`.`last_day_of_the_month`;
```

Add parameter

	subscription_id	customer_id	start_date	month_end_date
1	SUB001	101	2026-01-10T00:00:00.000+00:00	2026-01-31
2	SUB002	102	2026-02-15T00:00:00.000+00:00	2026-02-28
3	SUB003	103	2026-03-20T00:00:00.000+00:00	2026-03-31
4	SUB004	104	2026-04-25T00:00:00.000+00:00	2026-04-30
5	SUB005	105	2026-05-30T00:00:00.000+00:00	2026-05-31

Q15 This syntax extracts the first day of the month

```
select * from `retail`.`default`.`campaign_sends` limit 100;
```

```
SELECT
  send_id,
  customer_id,
  send_date,
  DATE_TRUNC('month', send_date) AS month_start_date
FROM `retail`.`default`.`campaign_sends`
```

Add parameter

	send_id	customer_id	send_date	month_start_date
1	C001	101	2026-01-12T00:00:00.000+00:00	2026-01-01T00:00:00.000+00:00
2	C002	102	2026-02-18T00:00:00.000+00:00	2026-02-01T00:00:00.000+00:00
3	C003	103	2026-03-22T00:00:00.000+00:00	2026-03-01T00:00:00.000+00:00
4	C004	104	2026-04-09T00:00:00.000+00:00	2026-04-01T00:00:00.000+00:00
5	C005	105	2026-05-27T00:00:00.000+00:00	2026-05-01T00:00:00.000+00:00

Q16 This syntax extracts, format the month and year

```
select * from `retail`.`default`.`invoice_dates` limit 100;
```

```
SELECT
  invoice_id,
  customer_id,
  invoice_date,
  DATE_FORMAT(invoice_date, 'MMMM yyyy') AS invoice_month_year
FROM `retail`.`default`.`invoice_dates`;
```

Add parameter

	invoice_id	customer_id	invoice_date	invoice_month_year
1	INV001	101	2026-01-05T00:00:00.000+00:00	January 2026
2	INV002	102	2026-02-10T00:00:00.000+00:00	February 2026
3	INV003	103	2026-03-15T00:00:00.000+00:00	March 2026
4	INV004	104	2026-04-20T00:00:00.000+00:00	April 2026
5	INV005	105	2026-05-25T00:00:00.000+00:00	May 2026

Q17 This syntax calculates customer's age in years using their date of birth and today's date.

```
select * from `retail`.`default`.`customer_birthdays` limit 100;
SELECT
  ... customer_id,
  ... customer_name,
  ... `date of birth`,
  ... TIMESTAMPDIFF(YEAR, `date of birth`, CURRENT_DATE()) AS customer_age
FROM `retail`.`default`.`customer_birthdays`;
```

Add parameter

Table ▾ + 🔍 🏠 📄

	1 2 3 customer_id	A B C customer_name	📅 date of birth	1 2 3 customer_age	
1	901	John	1988-05-10T00:00:00.000+00:...	38	
2	902	Mary	1990-08-20T00:00:00.000+00:...	35	
3	903	Peter	2002-03-15T00:00:00.000+00:...	24	
4	904	Sarah	1985-12-01T00:00:00.000+00:...	40	
5	905	Thabo	2000-07-30T00:00:00.000+00:...	25	

Q18 This syntax extracts weekdays vs weekend.

```
select * from `retail`.`default`.`week_vs_weekend` limit 100;
SELECT
  order_id,
  customer_id,
  `date of birth` AS order_date,
  DAYNAME(`date of birth`) AS day_name,
  CASE
    WHEN DAYNAME(`date of birth`) IN ('Saturday', 'Sunday') THEN 'Weekend'
    ELSE 'Weekday'
  END AS day_type
FROM `retail`.`default`.`week_vs_weekend`;
```

Add parameter

	order_id	customer_id	order_date	day_name	day_type
1	9001	101	2026-05-01T00:00:00.000+00:...	Fri	Weekday
2	9002	102	2026-05-02T00:00:00.000+00:...	Sat	Weekday
3	9003	103	2026-05-03T00:00:00.000+00:...	Sun	Weekday
4	9004	104	2026-05-04T00:00:00.000+00:...	Mon	Weekday
5	9005	105	2026-05-05T00:00:00.000+00:...	Tue	Weekday

Q19 This syntax extracts the quarterly transactions

```
select * from `database`.`default`.`quarterly_transaction` limit 100;
SELECT
  transaction_id,
  customer_id,
  `transaction date` AS transaction_date,
  QUARTER(`transaction date`) AS transaction_quarter,
  amount
FROM `database`.`default`.`quarterly_transaction`
```

Add parameter

	transaction_id	customer_id	transaction_date	transaction_quarter	amount
1	Q1001	101	2026-01-15T00:00:00.000+00:...	1	500
2	Q1002	102	2026-03-20T00:00:00.000+00:...	1	1200
3	Q1003	103	2026-04-10T00:00:00.000+00:...	2	800
4	Q1004	104	2026-07-05T00:00:00.000+00:...	3	1500
5	Q1005	105	2026-10-25T00:00:00.000+00:...	4	2000

Q20 This syntax extracts orders that are more than 30 days.

```
select * from `retail`.`default`.`recent_orders` limit 100;
SELECT
  order_id,
  customer_id,
  order_date,
  DATEDIFF(CURRENT_DATE(), order_date) AS days_since_order,
  amount
FROM `retail`.`default`.`recent_orders`
WHERE DATEDIFF(CURRENT_DATE(), order_date) > 30;
```

Add parameter

	order_id	customer_id	order_date	days_since_order	amount
1	1	101	2026-04-01T00:00:00.000+00:...	68	500
2	2	102	2026-04-15T00:00:00.000+00:...	54	1200
3	3	103	2026-05-01T00:00:00.000+00:...	38	800
4	5	105	2026-02-25T00:00:00.000+00:...	103	2000

The bonus question combines DATEDIFF, CASE, and CURRENT_DATE into a single customer segmentation query. Apply every concept from this guide in one question.

```
SELECT
  customer_id,
  `customer name` AS customer_name,
  `last purchase date` AS last_purchase_date,
  DATEDIFF(CURRENT_DATE(), `last purchase date`) AS days_since_last_purchase,
  CASE
    WHEN DATEDIFF(CURRENT_DATE(), `last purchase date`) <= 30
      THEN 'Active Customer'
    WHEN DATEDIFF(CURRENT_DATE(), `last purchase date`) BETWEEN 31 AND 90
      THEN 'At Risk Customer'
    ELSE 'Inactive Customer'
  END AS customer_segment
FROM `retail`.`default`.`customer_recency`;
```

Add parameter

	customer_id	customer_name	last_purchase_date	days_since_last_purchase	customer_segment
1	1001	John	2026-05-25T00:00:00.000+00:...	14	Active Customer
2	1002	Mary	2026-05-10T00:00:00.000+00:...	29	Active Customer
3	1003	Peter	2026-04-01T00:00:00.000+00:...	68	At Risk Customer
4	1004	Sarah	2026-02-15T00:00:00.000+00:...	113	Inactive Customer
5	1005	Thabo	2025-12-20T00:00:00.000+00:...	170	Inactive Customer

Conclusion: Date difference and case statements broad an interesting perspective when it comes to SQL. I have learned the is more you can extract from a simple value such a date.